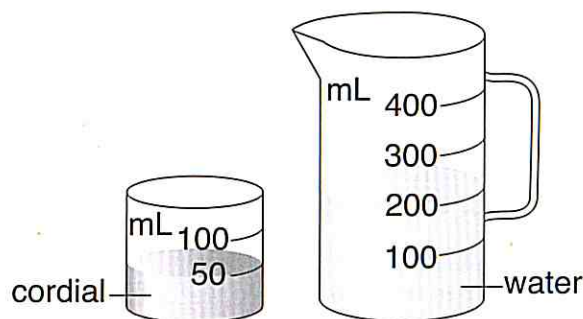


TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

(A) (B) (C) **(D)**



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)
- 11 (A) (B) (C) (D)
- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)



S

2012 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	B	B	D	C	C	B	B	D
2	B	B	D	B	B	D	A	C	C
3	A	A	D	A	D	B	C	B	C
4	C	C	C	B	C	B	A	C	C
5	C	D	B	C	B	C	C	B	D
6	B	C	D	C	C	C	C	A	C
7	A	A	C	D	B	B	B	A	B
8	B	A	B	B	D	A	D	A	A
9	C	B	B	B	B	C	C	D	A
10	D	C	C	A	C	B	A	D	D
11	C	D	D	A	B	D	B	C	D
12	D	D	C	A	C	C	B	C	D
13	A	C	D	B	B	D	A	B	B
14	B	A	C	D	D	D	B	A	C
15	B	A	B	A	D	C	B	C	C
16	A	B	A	A	B	D	C	D	B
17	B	A	B	C	C	B	D	C	C
18	C	A	A	B	A	B	A	B	C
19	D	C	A	B	C	B	A	A	B
20	A	B	B	C	C	A	D	C	B
21	D	D	C	A	A	A	B	B	A
22	B	B	B	B	A	A	C	D	D
23	D	D	A	D	C	D			



S

2012 Science Answer Keys

ICAS

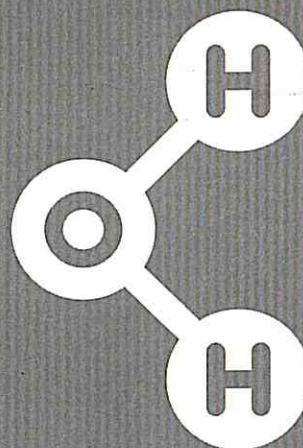
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			A	C	B	D	D	D	D
32			C	D	D	D	D	D	D
33			D	D	D	A	B	C	A
34			C	B	D	B	C	C	C



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**PAPER
B**



2012
ICAS

International Competitions
and Assessments for Schools

SCIENCE

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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

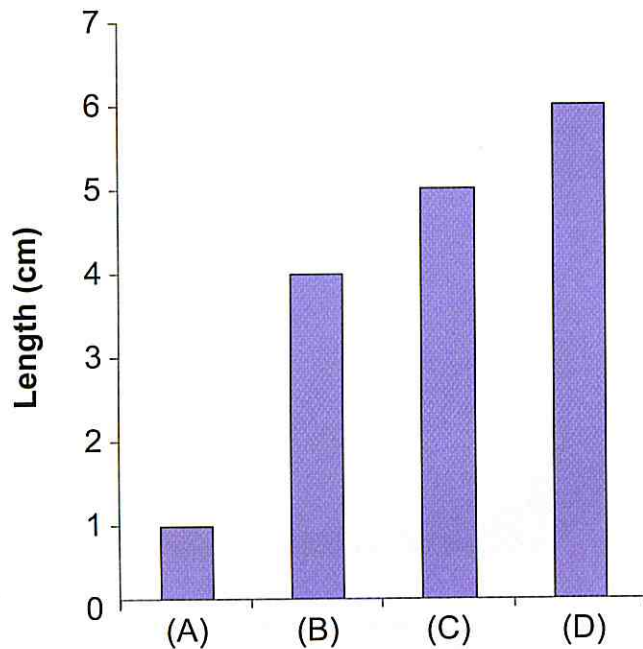
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Luca measured the length of a slug using a centimetre ruler.



Which column on the graph represents the length of the slug?



2. Allan was following the instructions to bake a cake. The recipe used symbols to show him what to do.

This symbol means 'add one heaped teaspoon of sugar'.



Which of the following symbols means 'add two heaped teaspoons of sugar'?

(A)



(B)



(C)



(D)

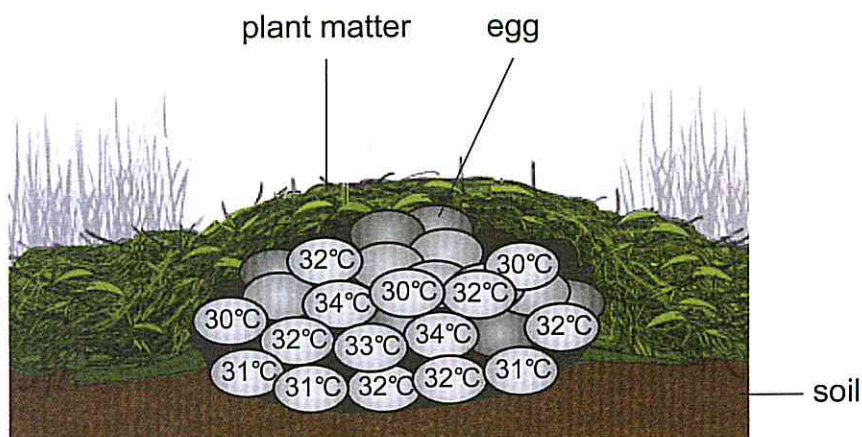


3. Female saltwater crocodiles lay their eggs in a nest in the soil and then cover the eggs with plant matter.

The temperature of each egg determines whether the egg will hatch a male crocodile or a female crocodile.

- Eggs with a temperature below 32 °C will always hatch female crocodiles.
- Eggs with a temperature of 32 °C will always hatch male crocodiles.
- Eggs with a temperature above 32 °C will always hatch female crocodiles.

The diagram below shows crocodile eggs in a nest. The temperature of some eggs is marked on them.



How many of the marked eggs will hatch as male crocodiles, and how many of the marked eggs will hatch as female crocodiles?

	Male crocodiles	Female crocodiles
(A)	6	9
(B)	6	6
(C)	9	6
(D)	12	3

4. Mary was growing a plant. She measured how tall the plant was at the end of each week, for seven weeks. The plant was 9 cm tall at the start of week 1.

Her results are shown in the table.

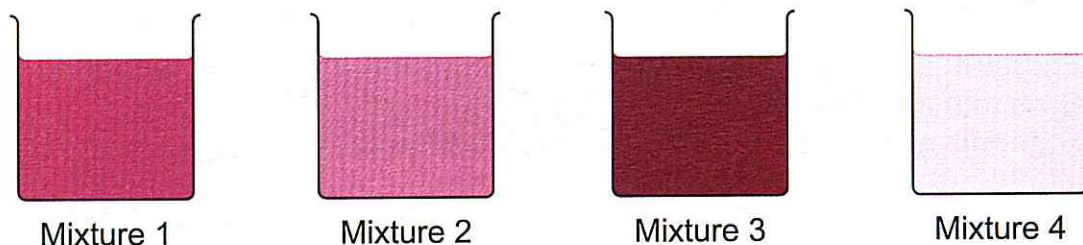
End of week	Height of plant (cm)
1	10
2	11
3	14
4	17
5	21
6	23
7	24

During which week did the plant grow the most?

- (A) week 1
- (B) week 4
- (C) week 5
- (D) week 7

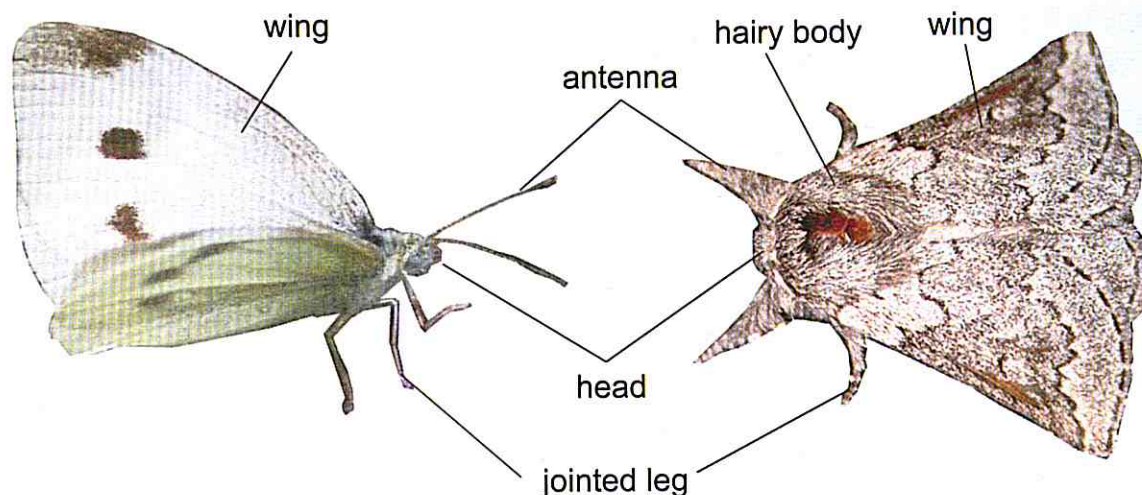
5. Emma made four mixtures by adding different amounts of food colouring to water. She used the same type of food colouring in every mixture.

The diagrams show the four mixtures she made.



Which option lists the mixtures from the one with the smallest amount of food colouring to the one with the largest amount of food colouring?

- (A) 3, 1, 2, 4
(B) 3, 2, 1, 4
(C) 4, 1, 2, 3
(D) 4, 2, 1, 3
6. The photographs show a butterfly and a moth at rest. The butterfly is viewed from the side. The moth is viewed from above. Some body parts are labelled.

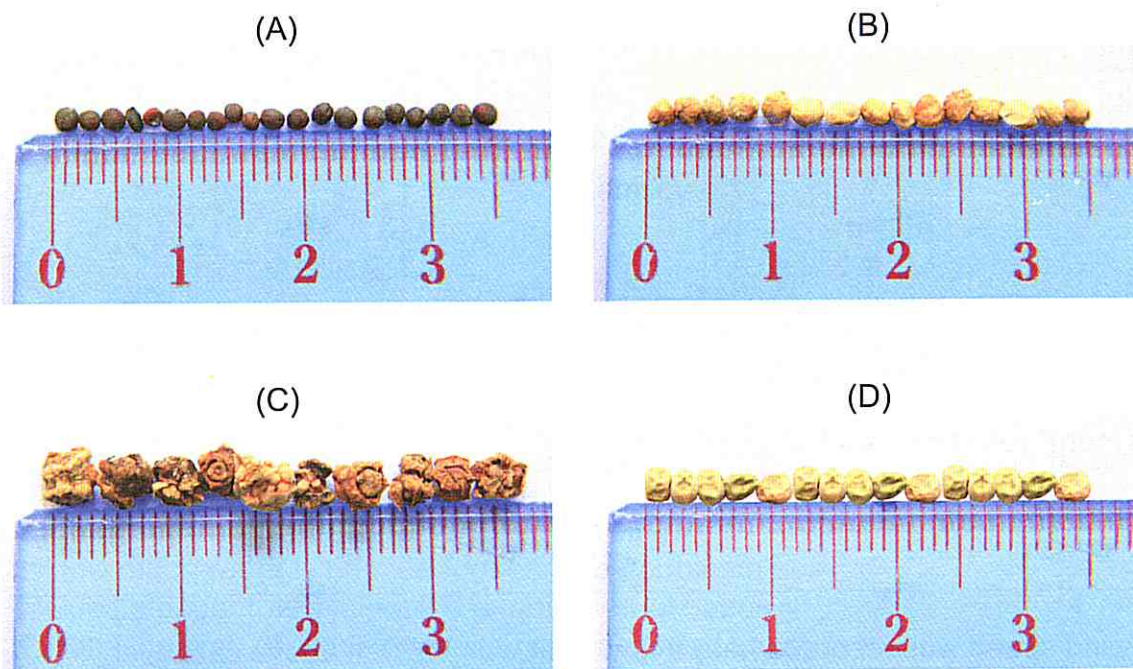


Which feature is common to both animals?

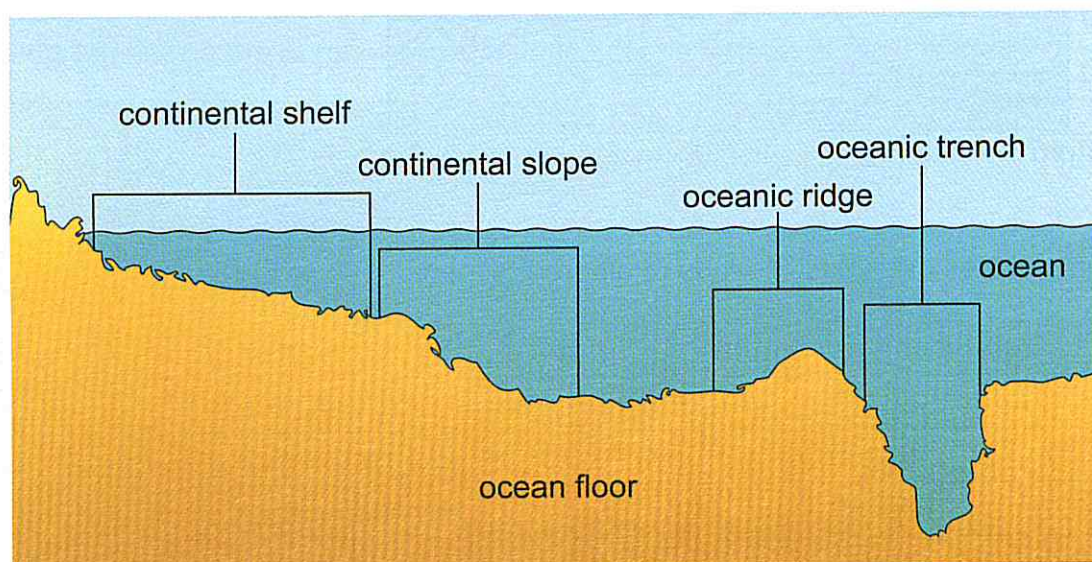
- (A) hairy body
(B) long thin antennae
(C) legs that are jointed
(D) position of wings when resting

7. The photographs show seeds from four different plants next to a ruler marked in centimetres.

Which seeds are the smallest?



8. Algae are aquatic plants that need a large amount of sunlight to survive.



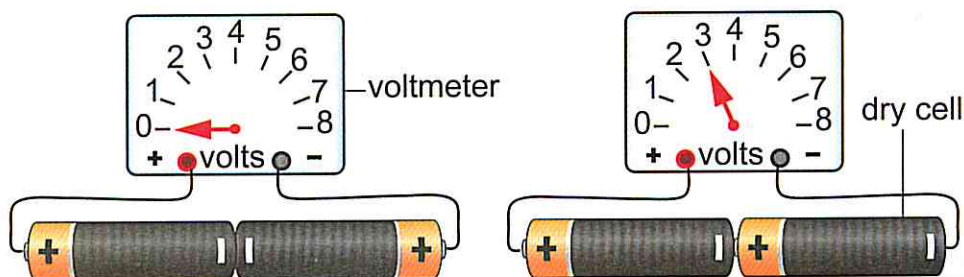
On which part of the ocean floor are algae most likely to survive?

- (A) continental shelf
- (B) continental slope
- (C) oceanic ridge
- (D) oceanic trench

9. A dry cell produces a voltage. This can be measured using a voltmeter.

When dry cells are joined, the total voltage depends on the number of cells and how they are connected.

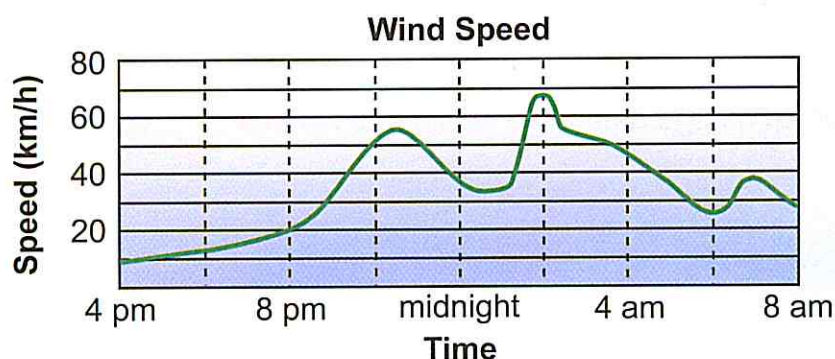
The diagrams show two ways that dry cells may be connected and the number of volts produced in each case.



How many volts does each dry cell produce?

- (A) 0 (B) $1\frac{1}{2}$ (C) 3 (D) $4\frac{1}{2}$

10. The wind speed was recorded near a town that was 1970 m above sea level.



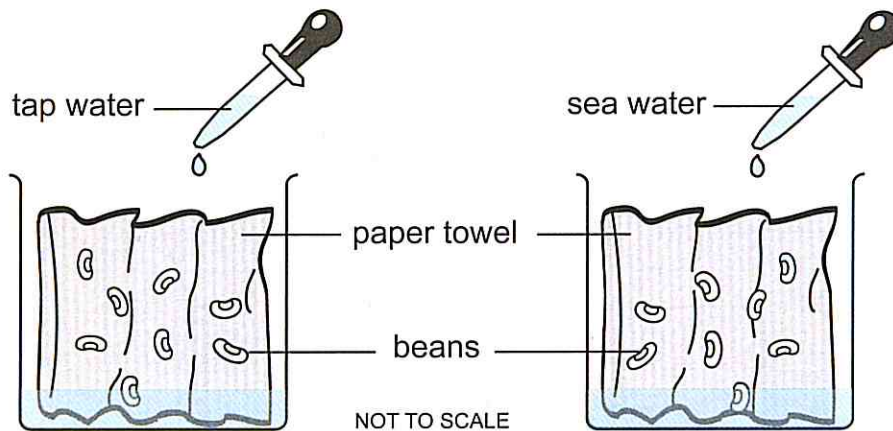
The table shows the types of winds of different speeds.

Speed of wind (km/h)	Type of wind
30–49	fresh breeze
50–59	moderate gale
60–69	fresh gale
70–84	strong gale

What type of wind was blowing at 2 am?

- (A) fresh breeze (B) moderate gale
(C) fresh gale (D) strong gale

11. Sarah saw this diagram of an experiment in a science book.



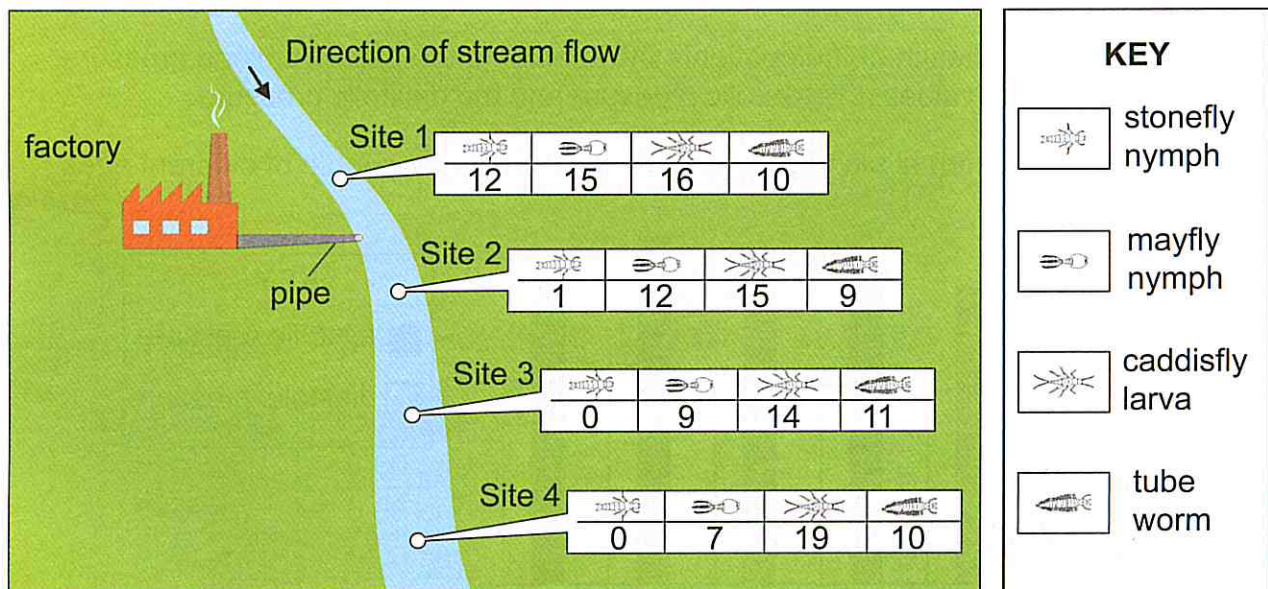
She decided to carry out the experiment herself.

Which of these questions will Sarah be able to answer from the results of her experiment?

- (A) Do beans germinate faster in groups of eight?
- (B) How much water is needed for beans to germinate?
- (C) Does a paper towel affect the germination of beans?
- (D) Does the type of water affect the germination of beans?

12. Small animals that spend their life under water can be good indicators of pollution. When a stream is polluted, some of these animals will no longer be able to live in the stream.

This diagram shows the number of animals found in a stream at Sites 1 to 4.

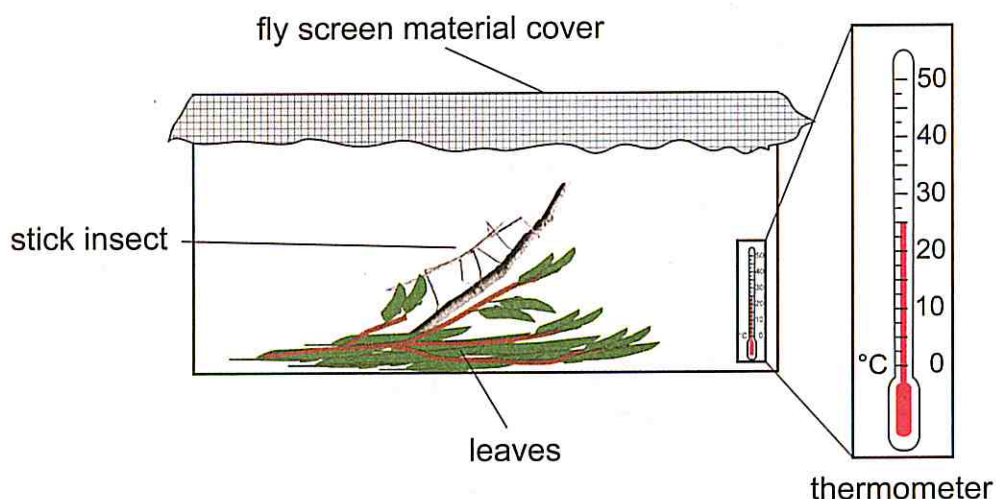


Which animal is the best indicator that the stream is polluted?

- (A) tube worm
- (B) mayfly nymph
- (C) caddisfly larva
- (D) stonefly nymph

For questions 13 and 14 use the information below.

The diagram shows a glass tank set up to keep stick insects. Stick insects only eat leaves.



13. What is the temperature (°C) in the tank?

- (A) 20 (B) 22 (C) 25 (D) 30

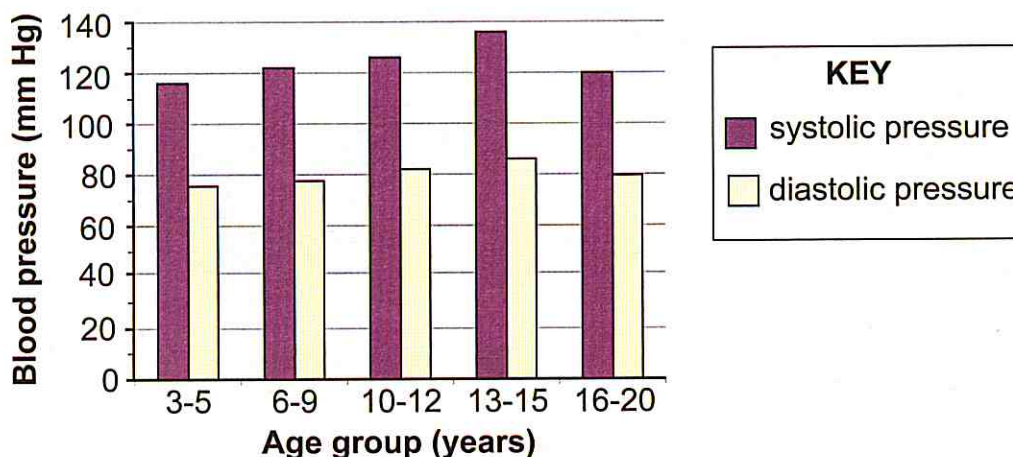
14. Jan placed ten stick insects into this tank. After a week she realised that the tank was too crowded with ten stick insects living in it.

Which observation would best support Jan's idea?

- (A) stick insects fighting each other (B) more moisture on the leaves
(C) stick insects eating leaves (D) a change in temperature

15. Doctors check the health of their patients by measuring the patient's blood pressure. They measure two values, the systolic pressure and the diastolic pressure.

The graph shows normal blood pressure measurements for people of different ages.

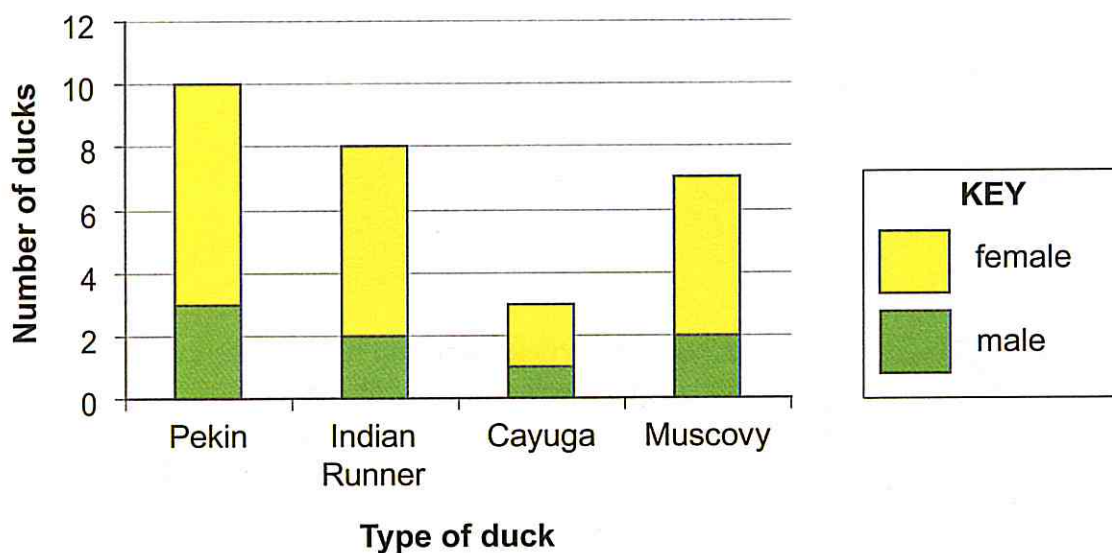


What is a normal diastolic blood pressure measurement (mm Hg) for an 8-year-old child?

- (A) 78 (B) 82 (C) 116 (D) 122

16. Madeline breeds ducks on her farm.

The graph gives information about the different types of duck she breeds.



How many male ducks does Madeline have?

- (A) 4 (B) 8 (C) 20 (D) 28

17. The table shows the times that the Sun rose and set during one week. The information for day 5 is missing.

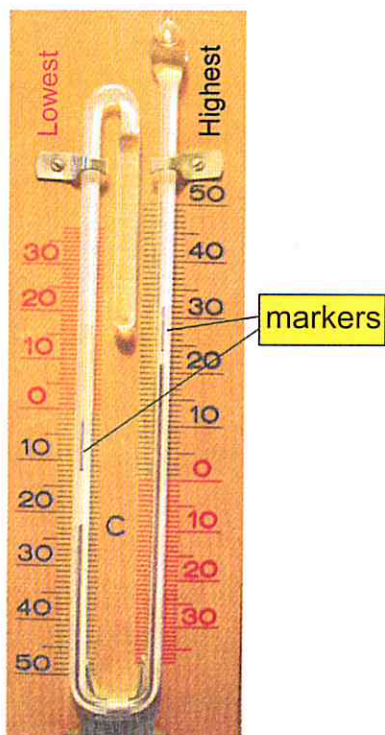
Day	Sun rose at	Sun set at
1	5:50 am	5:56 pm
2	5:48 am	5:57 pm
3	5:46 am	5:58 pm
4	5:44 am	5:59 pm
5		
6	5:40 am	6:01 pm
7	5:38 am	6:02 pm

Based on the table, at what times did the Sun rise and set on day 5?

	Sun rose at	Sun set at
(A)	5:42 am	6:00 pm
(B)	5:42 am	5:58 pm
(C)	5:46 am	6:00 pm
(D)	5:46 am	5:58 pm

18. A special type of thermometer shows the lowest temperature and the highest temperature reached since it was last reset. The thermometer does this by moving the markers upwards. Then the markers stay in position until they are reset.

The thermometer is reset by moving the markers back down until they both are touching the silver mercury in the U-tube again.



What were the lowest and highest temperatures reached since this thermometer was last reset?

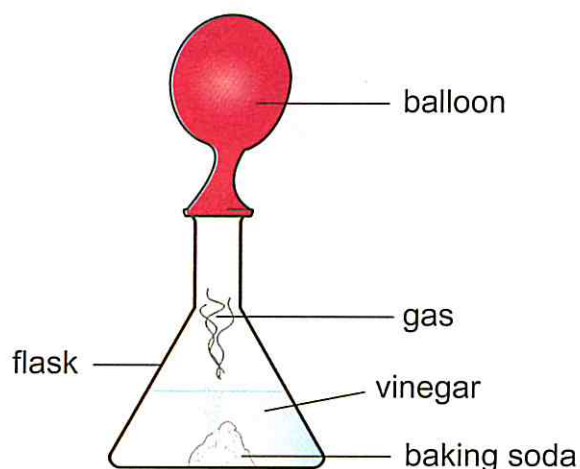
	Lowest temperature (°C)	Highest temperature (°C)
(A)	12	24
(B)	12	22
(C)	18	22
(D)	18	36

19. When vinegar is mixed with baking soda, bubbles of gas form. The gas can be trapped in a balloon.

Lina had a flask and a balloon which had a total mass of 250 g.

She put 15 g of baking soda and 55 g of vinegar into the flask and covered the mouth of the flask with the balloon before any gas could escape.

Bubbles started forming in the flask and the balloon expanded.



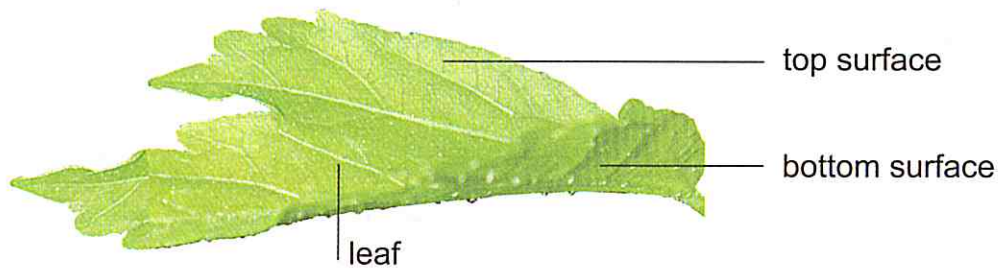
When the bubbling stopped, Lina weighed the flask with the balloon still attached without letting any gas escape.

What should be the total mass?

- (A) 250 g
 (B) 305 g
 (C) 320 g
 (D) more than 320 g

20. Leaves that have just been picked from healthy plants contain water. After a leaf is pulled off a plant, it loses water and dries up.

Water cannot escape from a leaf surface that has been coated with grease.



Bert carried out an investigation using four similar leaves taken from the same healthy plant.

He covered some surfaces of the leaves with grease:

- Leaf 1—all of its top surface
- Leaf 2—all of its bottom surface
- Leaf 3—all of its top and bottom surfaces
- Leaf 4—no grease on either surface.

Which question is Bert trying to answer with his investigation?

- (A) Does grease enter a leaf through its top surface?
- (B) From which surface does a leaf lose more water?
- (C) How quickly does a leaf with no grease on it dry up?
- (D) Do larger leaves lose more water than smaller leaves?

21. These photographs show what the Moon looked like on some nights during September. The photographs were taken from the same place on Earth, on the dates shown.

Sept 1



Sept 8



Sept 15



Sept 22



Sept 29



What would the Moon have looked like on September 26?

(A)



(B)



(C)



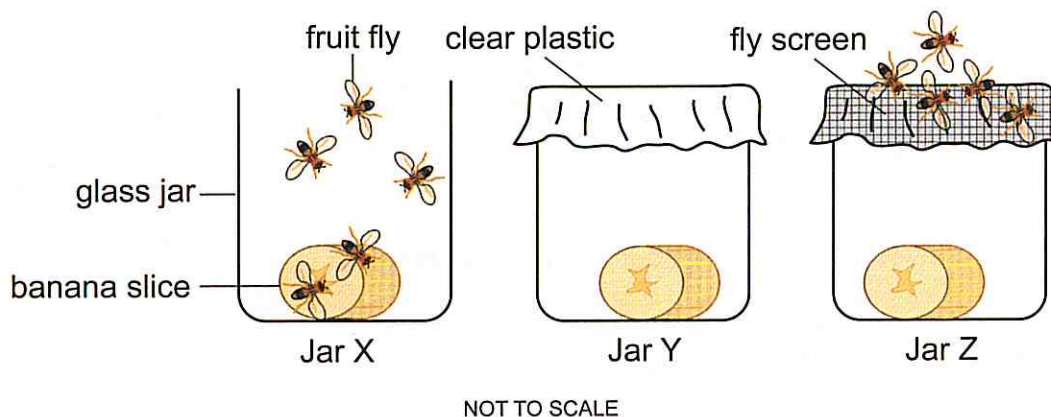
(D)



For questions 22 and 23 use the information below.

Nathan wanted to find out how fruit flies find bananas.

Nathan cut three pieces from one banana and placed them into three identical glass jars. He covered two of the jars with different materials. He placed all three jars together on a table outside. The diagrams show what Nathan observed a short time later.



22. Based on these observations, how do fruit flies find bananas?

- | | |
|------------------------------|------------------------|
| (A) by sight | (B) by smell |
| (C) by either sight or smell | (D) by smell and taste |

23. Which condition did Nathan have to keep the same for each jar to make his investigation a fair test?

- | | |
|-------------------------------|----------------------------------|
| (A) the type of cover | (B) the type of fruit fly |
| (C) the number of fruit flies | (D) the size of the banana slice |

24. Ben has four test tubes containing sugar solutions of different concentrations. The sugar concentrations are 5%, 10%, 15% and 20%.

Ben knows the following:

- test tube 4 contains the highest concentration
- test tube 1 does not have the lowest concentration
- test tube 2 contains a greater sugar concentration than test tube 1.

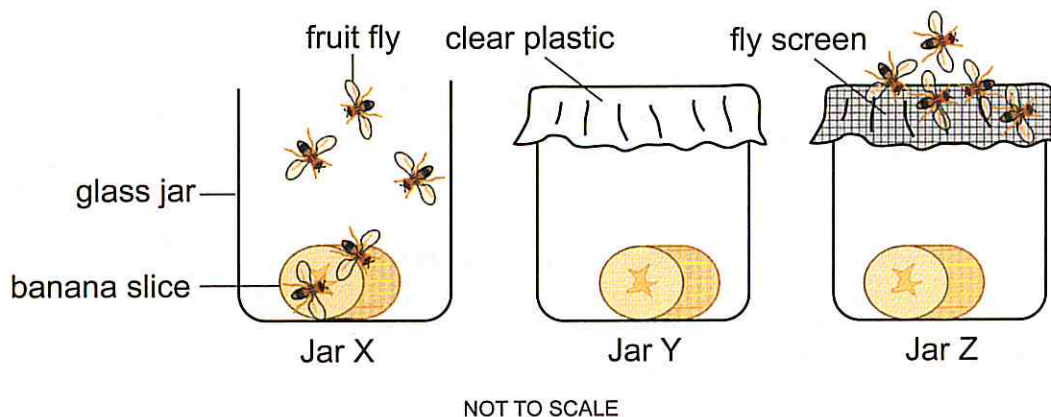
Which test tube contains the 15% sugar solution?

- | | | | |
|-------|-------|-------|-------|
| (A) 1 | (B) 2 | (C) 3 | (D) 4 |
|-------|-------|-------|-------|

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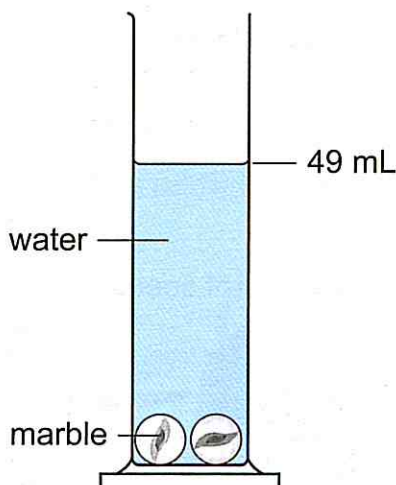
- test tube 4 contains the highest concentration
- test tube 1 does not have the lowest concentration
- test tube 2 contains a greater sugar concentration than test tube 1.

Which test tube contains the 15% sugar solution?

- | | | | |
|-------|-------|-------|-------|
| (A) 1 | (B) 2 | (C) 3 | (D) 4 |
|-------|-------|-------|-------|

25. Joel had a measuring cylinder with some water in it.

He added two identical marbles and the water level rose to 49 mL.



Joel then added a third identical marble and the water level rose to 51 mL.

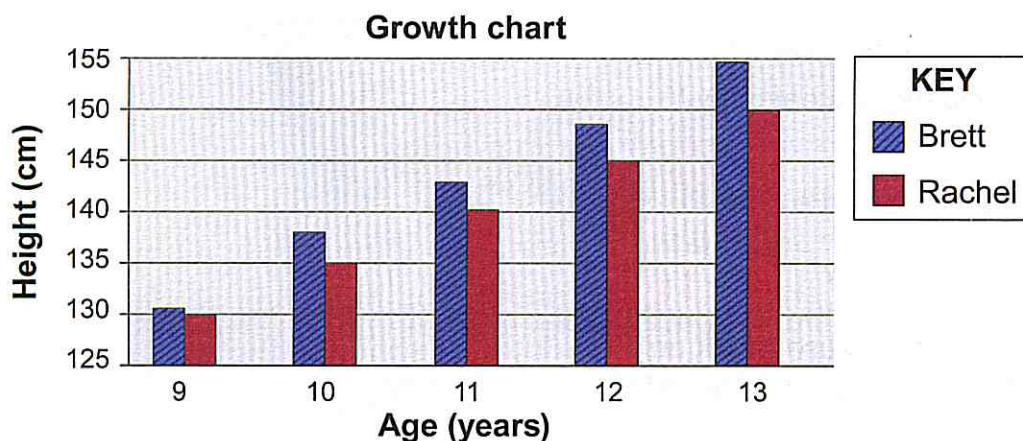
How much water did the cylinder contain just before Joel added the marbles?

- (A) 43 mL (B) 45 mL (C) 47 mL (D) 49 mL

For questions 26 and 27 use the information below.

Brett and Rachel had their heights measured on their birthdays over a number of years.

The growth chart shows this information.



26. How many centimetres did Brett grow between his 10th birthday and his 11th birthday?

- (A) 5 (B) 7 (C) 23 (D) 143

27. What conclusion can be drawn from the information in the growth chart?

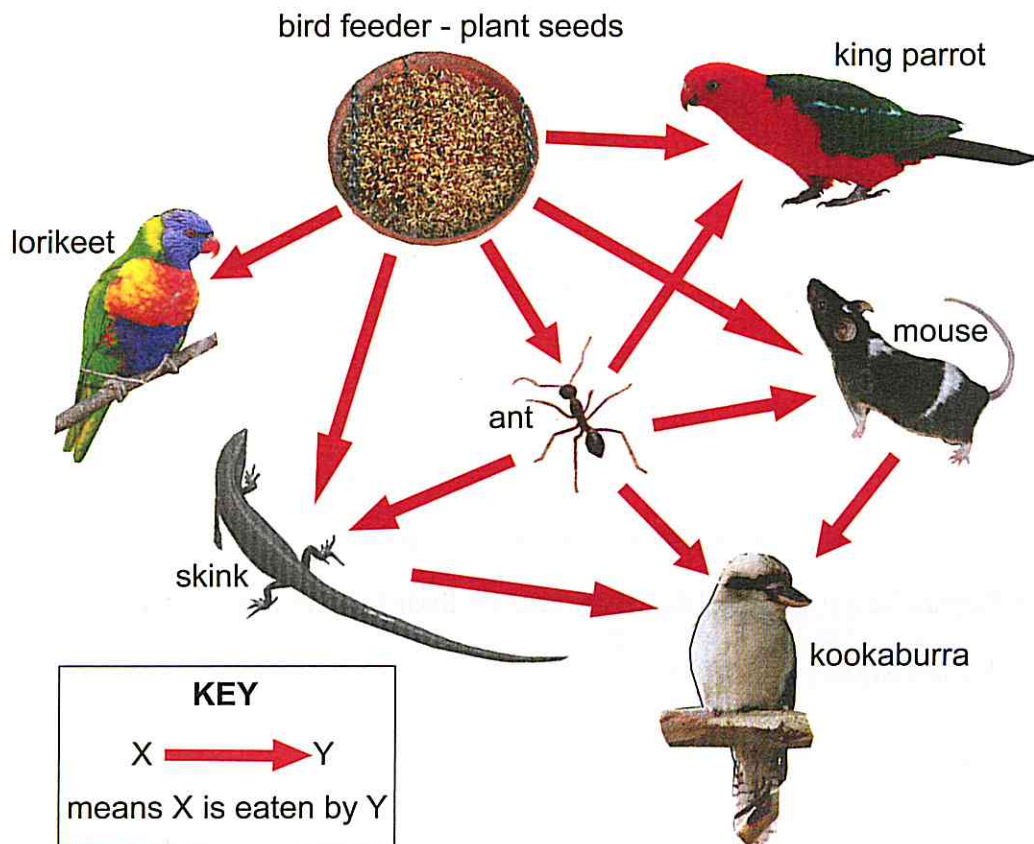
- (A) Brett is older than Rachel.
(B) Brett and Rachel have the same birthday.
(C) Brett was taller than Rachel on his 10th birthday.
(D) Brett grew faster than Rachel between the ages of 9 and 13.

For questions 28 and 29 use the information below.

There are different types of animals.

- Animals that only eat plant materials are called herbivores.
- Animals that only eat other animals are called carnivores.
- Animals that eat both plant materials and other animals are called omnivores.

The diagram shows the food web created when animals were attracted to a garden by a bird feeder.



28. Which of the animals are omnivores?

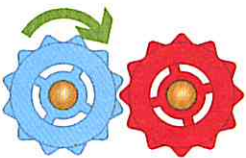
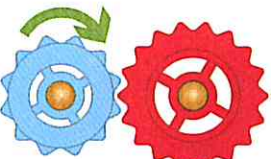
- (A) kookaburra only
- (B) ant, lorikeet only
- (C) king parrot, mouse, skink only
- (D) ant, king parrot, kookaburra, mouse, skink only

29. Which two animals from this food web are **not** in direct competition with each other for the same food?

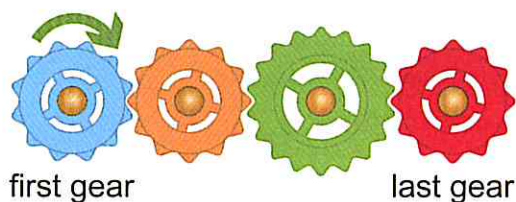
- (A) ant and skink
- (B) king parrot and skink
- (C) kookaburra and skink
- (D) kookaburra and lorikeet

30. Gears are wheels with teeth.

The table shows three sets of gears. In each set, the first gear makes 30 turns per minute in the direction shown. For each set of gears, the table shows the direction of rotation and speed of rotation of the last gear.

	Last gear	
	Direction of rotation	Speed of rotation
 first gear last gear		30 turns per minute
 first gear last gear		20 turns per minute
 first gear last gear		30 turns per minute

This diagram shows another set of gears. The first gear makes 30 turns per minute in the direction shown.



For this set of gears, what are the direction and the speed of rotation of the last gear?

	Direction of rotation	Speed of rotation
(A)		20 turns per minute
(B)		30 turns per minute
(C)		20 turns per minute
(D)		30 turns per minute

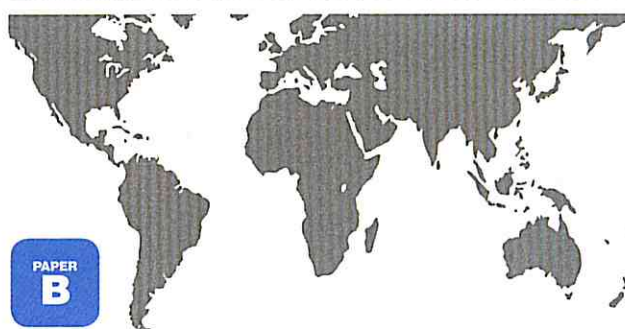
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Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 4
Singapore	Primary 3
South Africa	Grade 4



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